

Considering an **[Optimization Problem]** with assumptions $\mathcal{D} \neq \emptyset$, and optimal value p^* exists. The *Lagrangian* $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$ associated with the problem is defined as

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j h_j(x),$$

with $\text{dom } L = \mathcal{D} \times \mathbb{R}^m \times \mathbb{R}^p$. λ_i and ν_j are referred as *Lagrange multipliers* associated with respective inequality or equality constraint. The vector λ and ν are called *dual variables* or *Lagrange multiplier vectors* associated with the problem.

0.0.1 See also

0.0.2 References